Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Converting Forms (Completing the Square) **Date:** ___________

Foundation (Jalapeno)

Q1. Convert $x^2 + 6x + 8$ to vertex form.
(A) $(x+3)^2 - 1$
(B) $(x+3)^2 + 1$
(C) $(x+4)^2 - 8$
(D) $(x+2)^2 - 4$
(E) $(x+4)^2 + 8$

Q2. The equation of a planet's orbit is $y = x^2 - 4x - 3$. What is the vertex form?
(A) $(x-2)^2 - 7$
(B) $(x+2)^2 + 1$
(C) $(x-3)^2 - 2$
(D) $(x+1)^2 - 4$
(E) $(x-2)^2 - 5$

Q3. Complete the square: $x^2 + 2x$
(A) $(x+1)^2 - 1$
(B) $(x-1)^2 + 1$
(C) $(x+2)^2 - 4$
(D) $(x-2)^2 + 4$
(E) $(\mathrm{x} + 1)^2$

Q4. What is the vertex of the parabola $y = x^2 + 2x + 1$?
(A) $(-1, 0)$
(B) $(-1, 1)$
(C) $(-2, 0)$
(D) $(-2, -1)$
(E) (0, 1)

Q5. The equation of a lab experiment is $y = 2x^2 - 4x + 1$. What is the vertex?
(A) $(0, 1)$
(B) $(1, 0)$
(C) $(1, -1)$
(D) $(2, 0)$
(E) (1, 1)

Q6. Complete the square: $x^2 - 6x$
(A) $(x-3)^2 - 9$
(B) $(x-3)^2 + 3$
(C) $(x+3)^2 - 9$
(D) $(x+3)^2 + 9$
(E) $(x-3)^2 + 9$

Q7. What is the vertex form of $y = x^2 + 2x - 6$?
(A) $(x+1)^2 - 7$
(B) $(x-1)^2 + 5$
(C) $(x+1)^2 - 5$
(D) $(x+2)^2 - 2$
(E) $(x-2)^2 + 2$

Q8. The equation of a space mission is $y = -x^2 + 4x + 3$. What is the vertex form?
(A) $-(x-2)^2 + 7$
(B) $-(x+2)^2 - 1$
(C) $-(x-3)^2 + 2$
(D) $-(x-2)^2 + 5$
(E) $-(x+2)^2 + 1$

Open Ended Questions (FRQ)

Q9. Convert $x^2 + 8x + 12$ to vertex form and find the vertex.

Q10. Complete the square for $x^2 - 4x$ and write the equation in vertex form.

Chef's Tips

- Tip 1: Check for extraneous solutions
- Tip 2: Use the completing the square method
- Tip 3: Graph the parabola to verify the vertex